ENGR 225: Mechanics 1

Homework Coversheet

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DATE: _____

ASSIGNMENT: Homework 5

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

☐ None

■ Minor

☐ Major

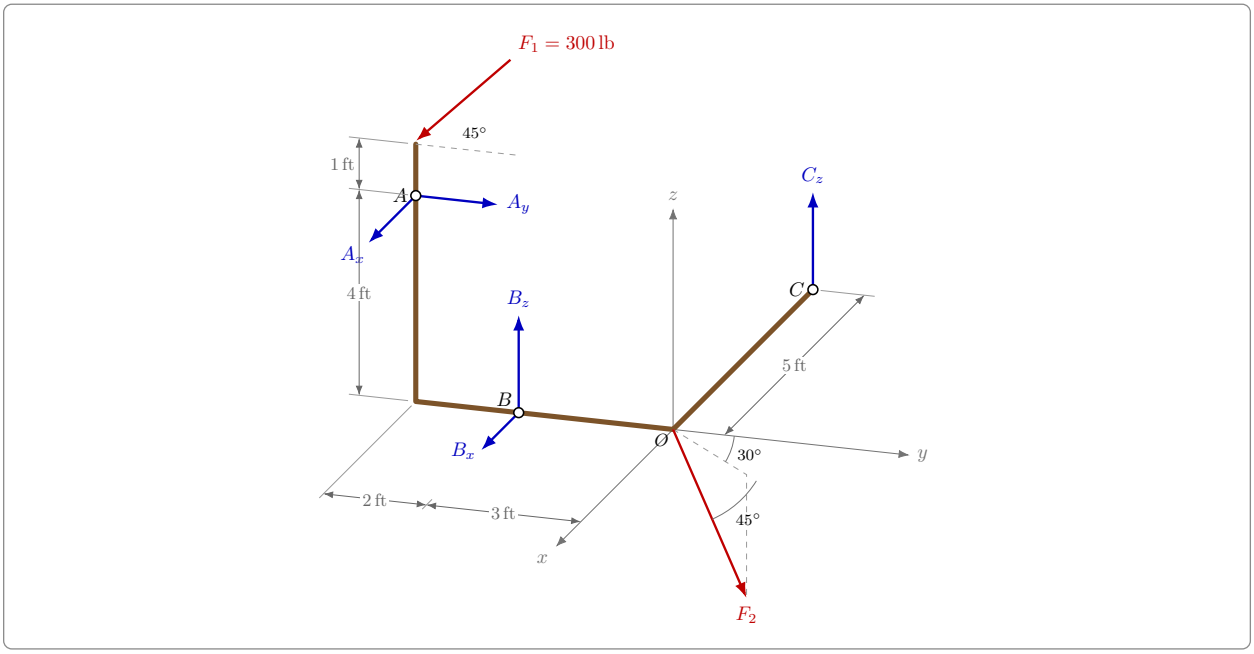
☐ Extensive

Mark the box that applies (e.g., replace ☐ with ■, or type an X beside it, before compiling/printing). Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

Problem 1

Given / Find: Rod ABC is held by smooth journal bearings at A, B, and C, which only exert forces on the rod. With $F_1 = 300\,\text{lb}$, find the magnitude of F_2 that makes the bearing reaction C_y zero.

Sketch:



Equations & Solution:
The coordinates (in ft) are $A(0,-5,4)$, $B(0,-3,0)$, and $C(-5,0,0)$, with F_1 applied at $(0,-5,5)$. Each bearing only resists force perpendicular to its shaft, so the unknowns are A_x and A_y at A, B_x and B_z at B, and C_z at C (C_y is set to zero). Together with F_2, that makes six unknowns for the six equilibrium equations.

$$\mathbf{F}_1 = 300(0,\,-\cos 45^\circ,\,-\sin 45^\circ) = (0,\,-212.1,\,-212.1)\ \text{lb}$$

$$\mathbf{F}_2 = F_2(\cos 45^\circ \sin 30^\circ,\ \cos 45^\circ \cos 30^\circ,\ -\sin 45^\circ) = F_2(0.3536,\ 0.6124,\ -0.7071)$$

I summed moments about O because F_2 acts at O and does not appear in any of the moment equations:

$$\sum F_x:\ A_x + B_x + 0.3536F_2 = 0 \qquad (1)$$

$$\sum F_y:\ A_y + 0.6124F_2 - 212.1 = 0 \qquad (2)$$

$$\sum F_z:\ B_z + C_z - 0.7071F_2 - 212.1 = 0 \qquad (3)$$

$$\sum M_x:\ -4A_y - 3B_z + 2121 = 0 \qquad (4)$$

$$\sum M_y:\ 4A_x + 5C_z = 0 \qquad (5)$$

$$\sum M_z:\ 5A_x + 3B_x = 0 \qquad (6)$$

I solved the system by substitution, writing each reaction in terms of F_2 and then plugging everything into (3).

From (5):

$$4A_x + 5C_z = 0 \ \Rightarrow\ A_x = -\tfrac{5}{4}C_z = -1.25\,C_z$$

From (6):

$$5A_x + 3B_x = 0 \ \Rightarrow\ B_x = -\tfrac{5}{3}A_x = -\tfrac{5}{3}(-1.25\,C_z) = 2.083\,C_z$$

Into (1):

$$\begin{aligned} A_x + B_x + 0.3536F_2 &= 0 \\ -1.25\,C_z + 2.083\,C_z + 0.3536F_2 &= 0 \\ 0.8333\,C_z &= -0.3536F_2 \\ C_z &= -0.4243F_2 \end{aligned}$$

From (2):

$$A_y = 212.1 - 0.6124F_2$$

Into (4):

$$-4A_y - 3B_z + 2121 = 0 \;\Rightarrow\; B_z = \frac{2121 - 4A_y}{3}$$

$$B_z = \frac{2121 - 4(212.1 - 0.6124F_2)}{3} = \frac{2121 - 848.4 + 2.4496F_2}{3}$$

$$B_z = 424.2 + 0.8165F_2$$

Into (3):

$$\begin{aligned} B_z + C_z - 0.7071F_2 - 212.1 &= 0 \\ (424.2 + 0.8165F_2) + (-0.4243F_2) - 0.7071F_2 - 212.1 &= 0 \\ 212.1 + (0.8165 - 0.4243 - 0.7071)F_2 &= 0 \\ 212.1 - 0.3149F_2 &= 0 \\ F_2 &= \frac{212.1}{0.3149} = 673.5\,\text{lb} \end{aligned}$$

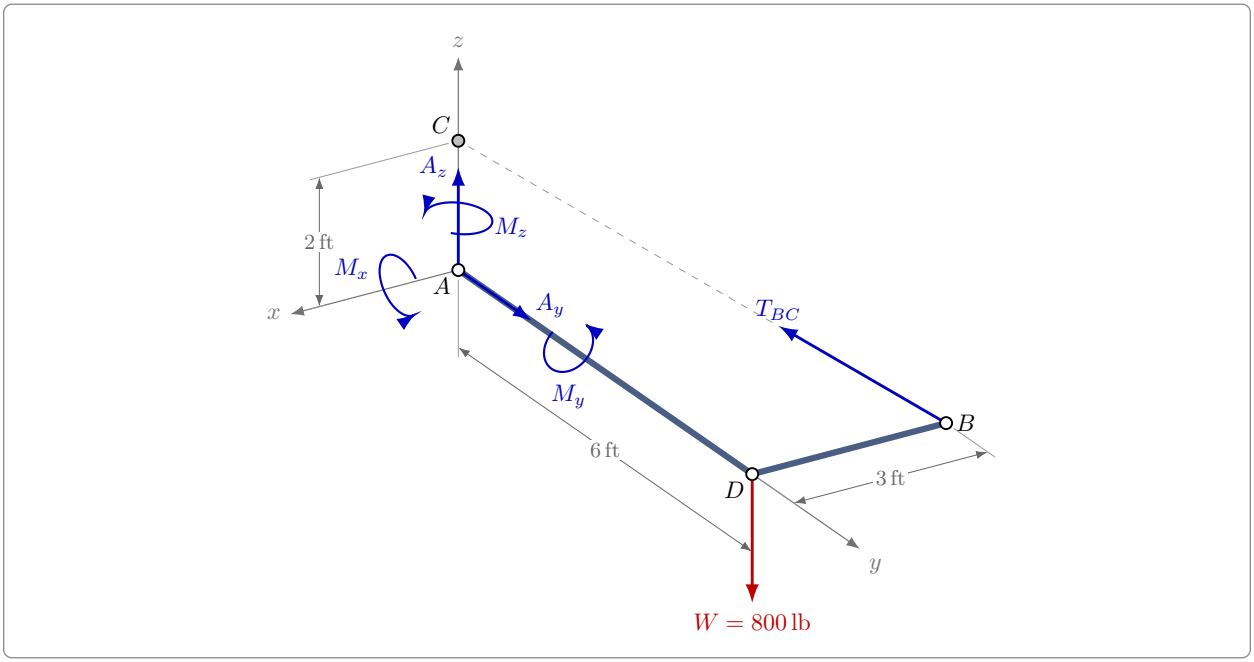
Answer

$F_2 = 674\,\text{lb}$

Problem 2

Given / Find: Member AB is supported by cable BC and by a square rod at A that fits loosely through the joint. Find the reactions at A and the cable tension T_{BC} needed to hold the 800 lb cylinder in equilibrium.

Sketch:



Equations & Solution:
Because the rod fits loosely, the joint can slide along it, so $A_x = 0$. The square cross-section still prevents rotation about all three axes, so A also has couple moments M_x, M_y, and M_z.

The coordinates (in ft) are $A(0,0,0)$, $D(0,6,0)$, $B(-3,6,0)$, and $C(0,0,2)$. The cable runs along $\overrightarrow{BC} = (3,-6,2)$, which has a length of 7 ft.

$$\mathbf{T}_{BC} = T_{BC}\left(\tfrac{3}{7}, -\tfrac{6}{7}, \tfrac{2}{7}\right), \qquad \mathbf{W} = (0,0,-800)\text{ lb at } D$$

$$\begin{aligned}
\textstyle\sum F_x: &\ \tfrac{3}{7}T_{BC} = 0 && \Rightarrow T_{BC} = 0\\
\textstyle\sum F_y: &\ A_y - \tfrac{6}{7}T_{BC} = 0 && \Rightarrow A_y = 0\\
\textstyle\sum F_z: &\ A_z + \tfrac{2}{7}T_{BC} - 800 = 0 && \Rightarrow A_z = 800\,\text{lb}
\end{aligned}$$

The cable is the only force on the member with an x-component, so its tension has to be zero.

With $T_{BC} = 0$, the weight is the only force with a moment about A: $\mathbf{r}_{AD} \times \mathbf{W} = (0,6,0) \times (0,0,-800) = (-4800,0,0)$ lb·ft.

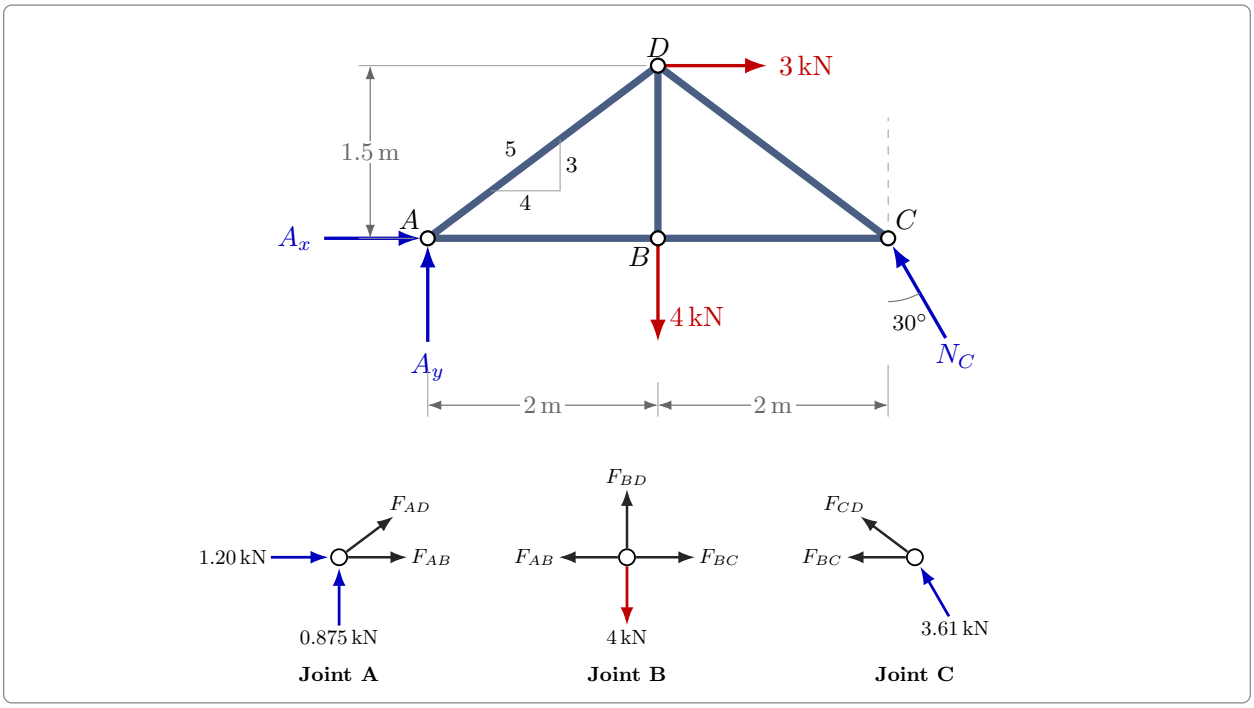
$$\begin{aligned}
\textstyle\sum M_x: &\ M_x - 4800 = 0 && \Rightarrow M_x = 4800\,\text{lb}\cdot\text{ft}\\
\textstyle\sum M_y: &\ M_y = 0, \qquad \textstyle\sum M_z: \ M_z = 0
\end{aligned}$$

Answer
$A_y = 0, \quad A_z = 800\,\text{lb}, \quad M_x = 4.80\times10^3\,\text{lb}\cdot\text{ft}, \quad M_y = M_z = 0, \quad T_{BC} = 0$

Problem 3

Given / Find: The truss has a pin at A and a roller at C on a surface inclined at $\theta = 30°$. Find the force in each member and state whether it is in tension or compression.

Sketch:



Equations & Solution:

The roller reaction N_C acts normal to the incline, which puts it 30° from vertical. Solving for the support reactions:

$$\sum M_A: \; N_C \cos 30°(4) - 4(2) - 3(1.5) = 0 \qquad \Rightarrow N_C = 3.608\,\text{kN}$$
$$\sum F_x: \; A_x + 3 - N_C \sin 30° = 0 \qquad \Rightarrow A_x = -1.196\,\text{kN}$$
$$\sum F_y: \; A_y - 4 + N_C \cos 30° = 0 \qquad \Rightarrow A_y = 0.875\,\text{kN}$$

Members AD and CD form 3-4-5 triangles, so their direction cosines are 0.8 and 0.6. Working through the joints:

$$A: \; \sum F_y = 0.875 + 0.6F_{AD} = 0 \qquad \Rightarrow F_{AD} = -1.458\,\text{kN}$$
$$\sum F_x = -1.196 + F_{AB} + 0.8F_{AD} = 0 \qquad \Rightarrow F_{AB} = 2.363\,\text{kN}$$
$$B: \; \sum F_x = F_{BC} - F_{AB} = 0 \qquad \Rightarrow F_{BC} = 2.363\,\text{kN}$$
$$\sum F_y = F_{BD} - 4 = 0 \qquad \Rightarrow F_{BD} = 4.000\,\text{kN}$$
$$C: \; \sum F_y = 3.125 + 0.6F_{CD} = 0 \qquad \Rightarrow F_{CD} = -5.208\,\text{kN}$$

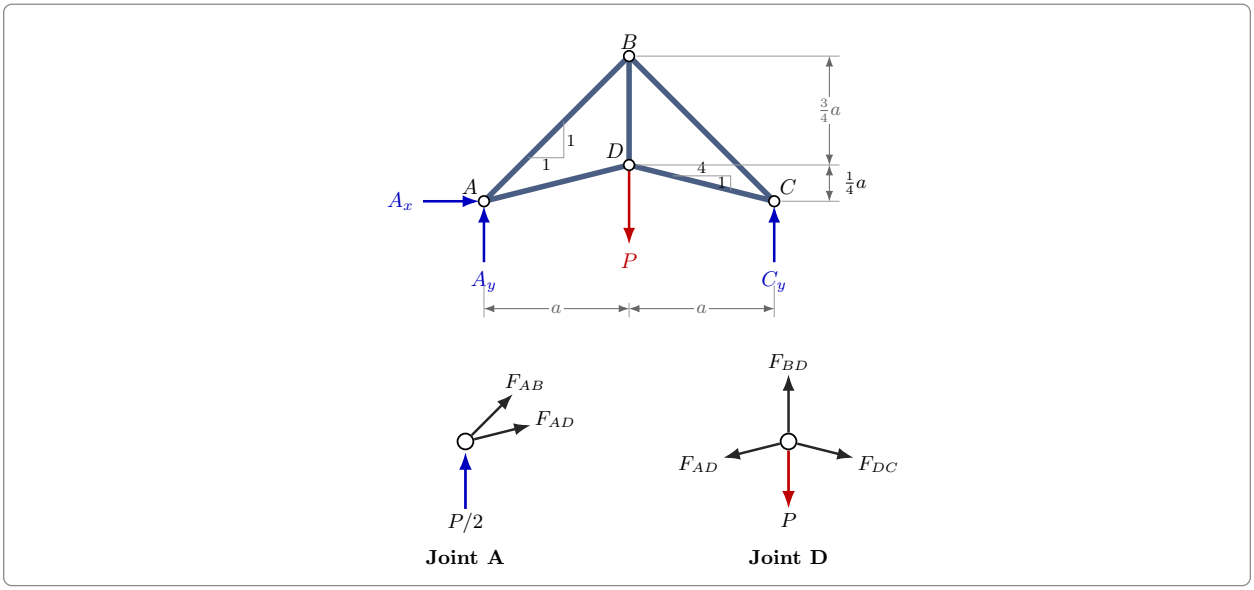
Answer

$F_{AB} = 2.36\,\text{kN}$ (T), $F_{AD} = 1.46\,\text{kN}$ (C), $F_{BC} = 2.36\,\text{kN}$ (T), $F_{BD} = 4.00\,\text{kN}$ (T), $F_{CD} = 5.21\,\text{kN}$ (C)

Problem 4

Given / Find: Members AB and BC can each support a maximum compressive force of 800 lb, and members AD, DC, and BD can each support a maximum tensile force of 2000 lb. With $a = 6$ ft, find the greatest load P the truss can carry.

Sketch:



Equations & Solution:

Member AB is at $45°$ and member AD is at $\varphi = \tan^{-1}(1/4) = 14.04°$. The truss and load are symmetric, so each support carries half the load ($A_y = C_y = P/2$), and $A_x = 0$ because there are no horizontal loads.

$$A: \ \sum F_x = F_{AB}\cos 45° + F_{AD}\cos\varphi = 0 \qquad \Rightarrow F_{AD} = -0.7289F_{AB}$$

$$\sum F_y = \tfrac{P}{2} + F_{AB}\sin 45° + F_{AD}\sin\varphi = 0 \qquad \Rightarrow F_{AB} = -0.9428P, \quad F_{AD} = 0.6872P$$

$$D: \ \sum F_y = F_{BD} - 2F_{AD}\sin\varphi - P = 0 \qquad \Rightarrow F_{BD} = 1.333P$$

By symmetry, $F_{BC} = F_{AB}$ and $F_{DC} = F_{AD}$. Setting each member force equal to its limit gives the largest P that member allows:

Member	Force	Limit	Max. P
AB, BC (C)	$0.9428P$	800 lb	848.5 lb
AD, DC (T)	$0.6872P$	2000 lb	2910 lb
BD (T)	$1.333P$	2000 lb	1500 lb

The smallest of these is 848.5 lb, from members AB and BC, so they reach their limit first. With P rounded down to 848 lb, the force in AB and BC is $0.9428(848) = 799.5$ lb, which stays under the 800 lb limit.

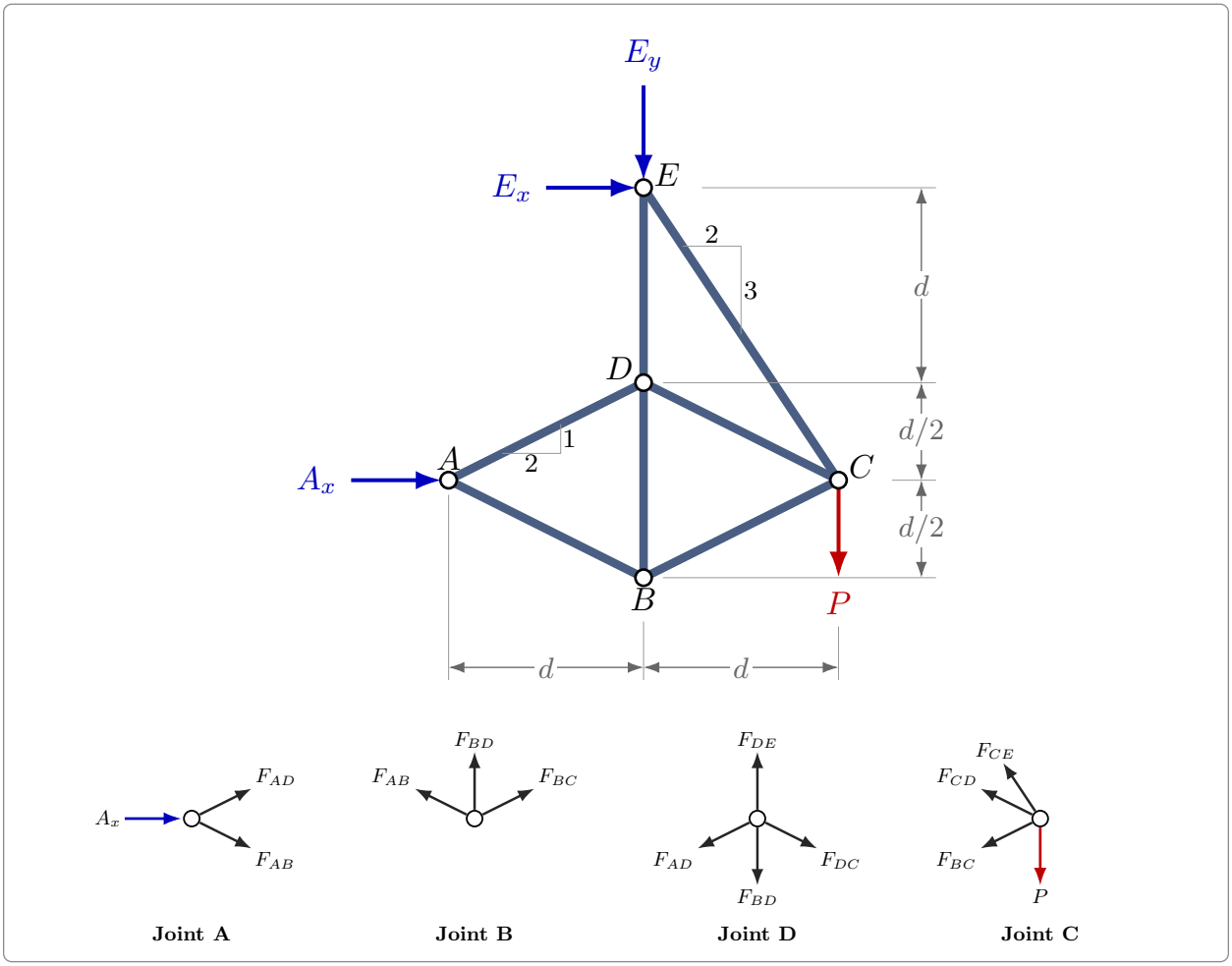
Answer

$P_{\max} = 848$ lb

Problem 5

Given / Find: The maximum allowable tensile force in any member is 5 kN, and the maximum allowable compressive force is 3 kN. With $d = 2\,\text{m}$, find the maximum load P that can be applied to the truss.

Sketch:



Equations & Solution:

E is a pin, so it has reactions E_x and E_y. A is a roller against the wall, so it can only push horizontally, which means $A_y = 0$.

Members AD, AB, DC, and BC have a 1:2 slope (direction cosines $\frac{2}{\sqrt{5}}$ and $\frac{1}{\sqrt{5}}$), and EC has a 3:2 slope (direction cosines $\frac{2}{\sqrt{13}}$ and $\frac{3}{\sqrt{13}}$).

Working through joints A, B, and D shows that DE is a zero-force member:

$$\begin{aligned}
A:\ &\textstyle\sum F_y = \frac{1}{\sqrt{5}}(F_{AD} - F_{AB}) = 0 && \Rightarrow F_{AD} = F_{AB} \\
B:\ &\textstyle\sum F_x = \frac{2}{\sqrt{5}}(F_{BC} - F_{AB}) = 0 && \Rightarrow F_{BC} = F_{AB} \\
&\textstyle\sum F_y = F_{BD} + \frac{1}{\sqrt{5}}(F_{AB} + F_{BC}) = 0 && \Rightarrow F_{BD} = -\frac{2}{\sqrt{5}}F_{AB} \\
D:\ &\textstyle\sum F_x = \frac{2}{\sqrt{5}}(F_{DC} - F_{AD}) = 0 && \Rightarrow F_{DC} = F_{AD} = F_{AB} \\
&\textstyle\sum F_y = F_{DE} - F_{BD} - \frac{1}{\sqrt{5}}(F_{AD} + F_{DC}) = 0 && \Rightarrow F_{DE} = -\frac{2}{\sqrt{5}}F_{AB} + \frac{2}{\sqrt{5}}F_{AB} = 0
\end{aligned}$$

Everything is now in terms of F_{AB}, so joint C only has two unknowns, F_{CE} and F_{AB}:

$$\begin{aligned}
&\textstyle\sum F_y = \frac{3}{\sqrt{13}}F_{CE} + \frac{1}{\sqrt{5}}(F_{CD} - F_{BC}) - P = 0 && \Rightarrow F_{CE} = \frac{\sqrt{13}}{3}P = 1.202P \\
&\textstyle\sum F_x = -\frac{2}{\sqrt{13}}F_{CE} - \frac{2}{\sqrt{5}}(F_{CD} + F_{BC}) = 0 && \Rightarrow F_{AB} = -0.3727P
\end{aligned}$$

This gives $F_{AD} = F_{AB} = F_{DC} = F_{BC} = 0.3727P$ (C), $F_{BD} = 0.3333P$ (T), $F_{CE} = 1.202P$ (T), and $F_{DE} = 0$. Setting each member force equal to its limit:

Member	**Force**	**Limit**	**Max.** P
CE (T)	$1.202P$	5 kN	4.160 kN
BD (T)	$0.3333P$	5 kN	15.0 kN
AD, AB, DC, BC (C)	$0.3727P$	3 kN	8.05 kN

Member CE reaches its 5 kN tension limit first, so it sets the maximum load.

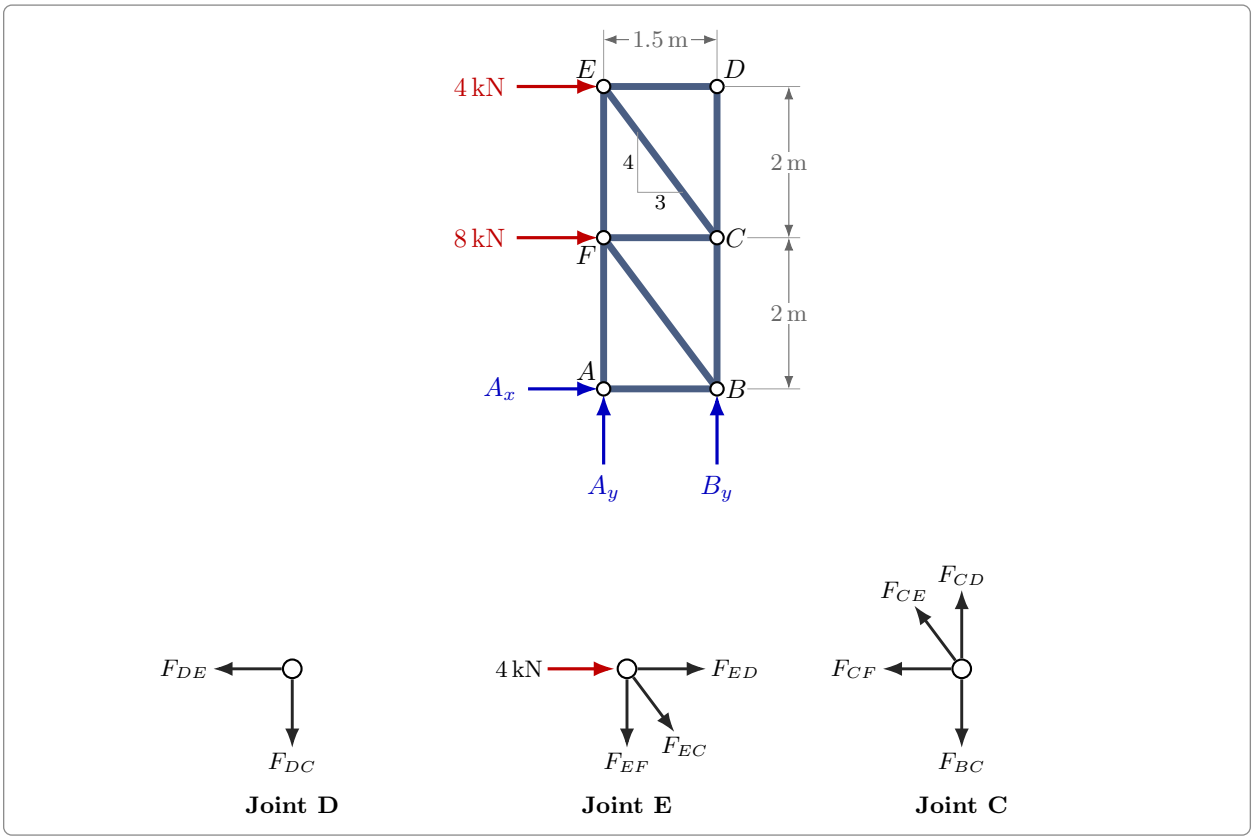
Answer

$P_{\max} = 4.16\,\text{kN}$

Problem 6

Given / Find: Determine the forces in members EF, CF, and BC and state whether they are in tension or compression.

Sketch:



Equations & Solution:
Joint D has no load and only two members, and they are not collinear, so DE and DC are both zero-force members. Starting at joint E means the support reactions are not needed.

Member EC spans 1.5 m horizontally and 2 m vertically, a 3-4-5 triangle, so its direction from E to C is $(0.6, -0.8)$.

$$
\begin{aligned}
E: \ \textstyle\sum F_x &= 4 + 0.6F_{EC} = 0 && \Rightarrow F_{EC} = -6.667\,\text{kN} \\
\textstyle\sum F_y &= -F_{EF} - 0.8F_{EC} = 0 && \Rightarrow F_{EF} = 5.333\,\text{kN} \\
C: \ \textstyle\sum F_x &= -0.6F_{CE} - F_{CF} = 0 && \Rightarrow F_{CF} = 4.000\,\text{kN} \\
\textstyle\sum F_y &= 0.8F_{CE} - F_{BC} = 0 && \Rightarrow F_{BC} = -5.333\,\text{kN}
\end{aligned}
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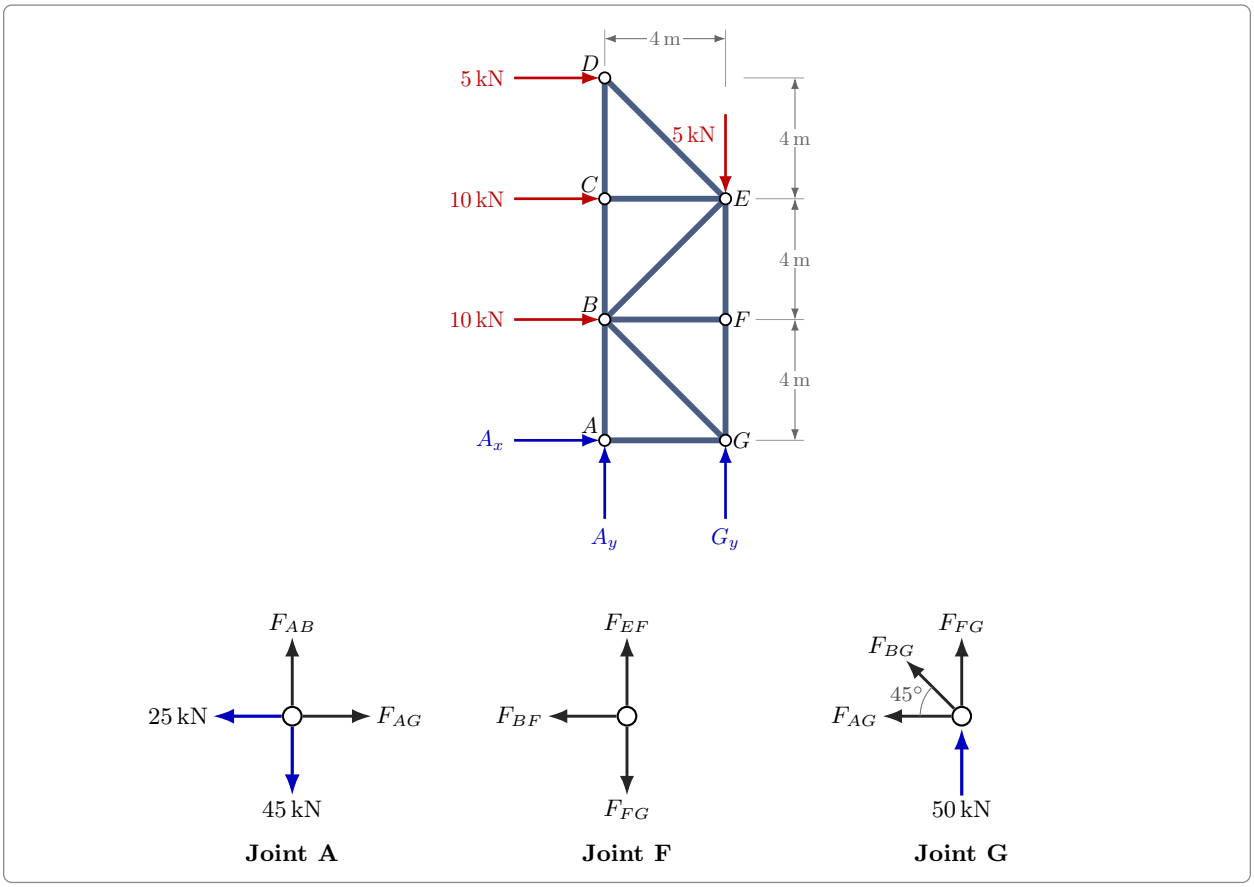
Answer

$F_{EF} = 5.33\,\text{kN}$ (T), $F_{CF} = 4.00\,\text{kN}$ (T), $F_{BC} = 5.33\,\text{kN}$ (C)

Problem 7

Given / Find: The truss has a pin at A and a roller at G. Determine the forces in members BF, BG, and AB and state whether they are in tension or compression.

Sketch:



Equations & Solution:

Solving for the support reactions:

$$\sum M_A: \; 4G_y - 5(12) - 10(8) - 10(4) - 5(4) = 0 \qquad \Rightarrow G_y = 50.0\,\text{kN}$$
$$\sum F_x: \; A_x + 25 = 0 \qquad \Rightarrow A_x = -25.0\,\text{kN}$$
$$\sum F_y: \; A_y + 50 - 5 = 0 \qquad \Rightarrow A_y = -45.0\,\text{kN}$$

At joint F, members EF and FG are collinear and there is no external load, so BF is a zero-force member and $F_{BF} = 0$. The other two forces come from joints A and G:

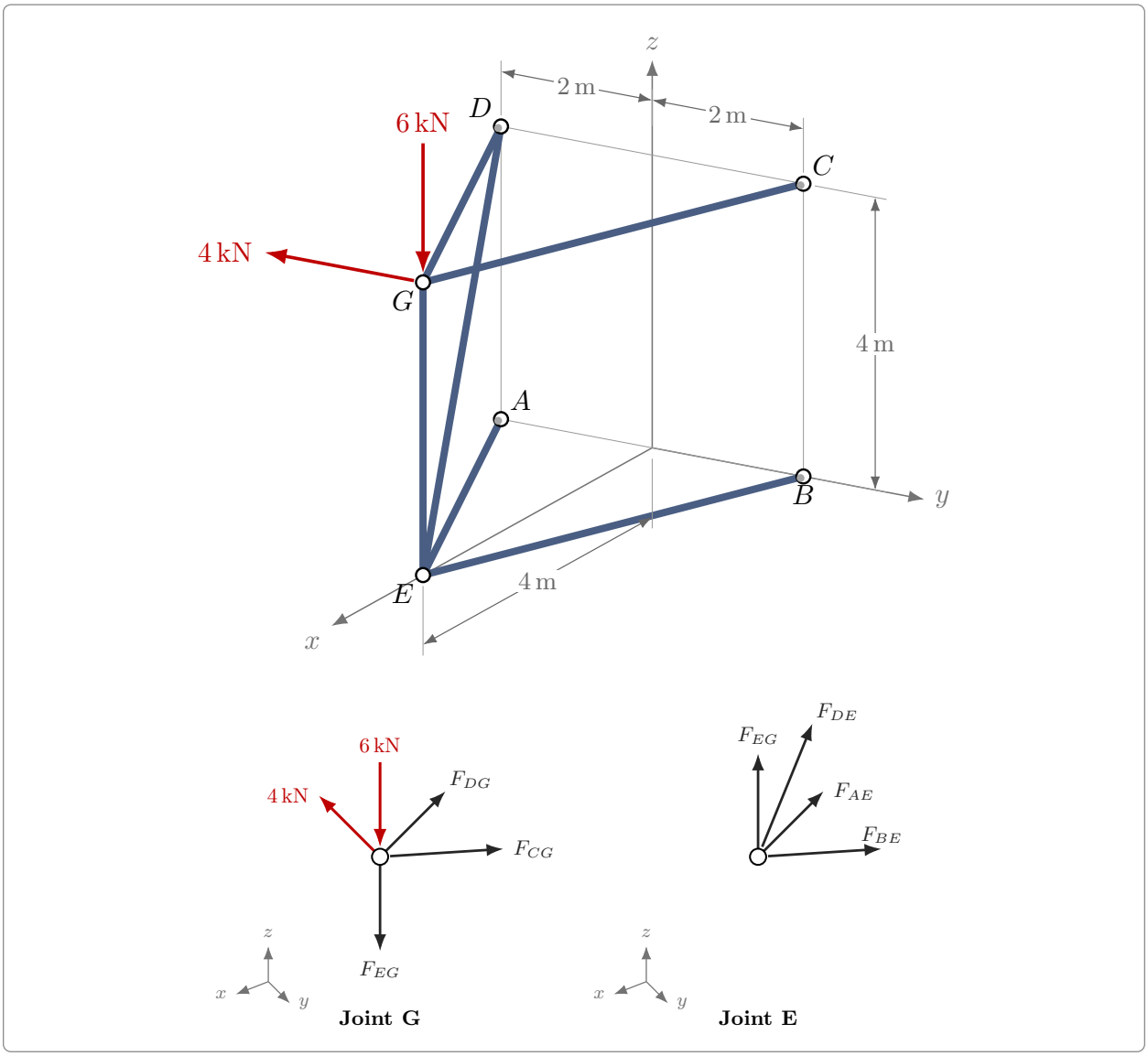
$$A: \; \sum F_y = -45 + F_{AB} = 0 \qquad \Rightarrow F_{AB} = 45.0\,\text{kN}$$
$$\sum F_x = -25 + F_{AG} = 0 \qquad \Rightarrow F_{AG} = 25.0\,\text{kN}$$
$$G: \; \sum F_x = -F_{AG} - F_{BG}\cos 45^\circ = 0 \qquad \Rightarrow F_{BG} = -35.36\,\text{kN}$$

Answer
$F_{AB} = 45.0\,\text{kN}$ (T), $\quad F_{BF} = 0$, $\quad F_{BG} = 35.4\,\text{kN}$ (C)

Problem 8

Given / Find: The space truss is supported by ball-and-socket joints at A, B, C, and D. Find the force in each member and state whether it is in tension or compression.

Sketch:



Equations & Solution:
The coordinates (in m) are $A(0,-2,0)$, $B(0,2,0)$, $C(0,2,4)$, $D(0,-2,4)$, $E(4,0,0)$, and $G(4,0,4)$. The joint FBDs are drawn from a rotated view so the members at E do not overlap. Joint G only has three unknown member forces, so I solved it first and then moved to E.

Joint G: the unit vectors from G toward E, C, and D are $(0,0,-1)$, $(-0.8944,\ 0.4472,\ 0)$, and $(-0.8944,\ -0.4472,\ 0)$. The applied load is $(0,-4,-6)$ kN.

$$\sum F_x = -0.8944(F_{CG} + F_{DG}) = 0 \qquad \Rightarrow F_{DG} = -F_{CG}$$
$$\sum F_y = 0.4472(F_{CG} - F_{DG}) - 4 = 0 \qquad \Rightarrow F_{CG} = 4.472\,\text{kN},\ F_{DG} = -4.472\,\text{kN}$$
$$\sum F_z = -F_{EG} - 6 = 0 \qquad \Rightarrow F_{EG} = -6.00\,\text{kN}$$

Joint E: the unit vectors from E toward A, B, and D are $(-0.8944,\ -0.4472,\ 0)$, $(-0.8944,\ 0.4472,\ 0)$, and $(-0.6667,\ -0.3333,\ 0.6667)$. There is no external load at E.

$$\sum F_z = F_{EG} + 0.6667F_{DE} = 0 \qquad \Rightarrow F_{DE} = 9.00\,\text{kN}$$
$$\sum F_x = -0.8944(F_{AE} + F_{BE}) - 0.6667(9) = 0 \qquad \Rightarrow F_{AE} + F_{BE} = -6.708$$
$$\sum F_y = 0.4472(F_{BE} - F_{AE}) - 0.3333(9) = 0 \qquad \Rightarrow F_{BE} - F_{AE} = 6.708$$

Solving the last two equations together gives $F_{BE} = 0$ and $F_{AE} = -6.708\,\text{kN}$.

Answer

$F_{AE} = 6.71\,\text{kN}$ (C), $F_{BE} = 0$, $F_{CG} = 4.47\,\text{kN}$ (T), $F_{DE} = 9.00\,\text{kN}$ (T), $F_{DG} = 4.47\,\text{kN}$ (C), $F_{EG} = 6.00\,\text{kN}$ (C)